

A Comprehensive Review on Optimization of Wire Electrical Discharge Machining (WEDM) Parameters and Performance Characteristics

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Abstract

Wire Electrical Discharge Machining (WEDM) is a widely used non-traditional machining process for producing complex and precise components from electrically conductive materials, especially hard-to-machine alloys and tool steels. The process operates on the principle of thermo-electric spark erosion, where material is removed by a series of controlled electrical discharges between a continuously moving wire electrode and the workpiece. Due to its high accuracy, excellent surface finish, and ability to machine intricate geometries, WEDM has become an essential technique in modern manufacturing industries.

However, the performance of the WEDM process is highly influenced by various machining parameters such as pulse-on time, pulse-off time, discharge current, wire feed, and wire tension. Proper selection and optimization of these parameters are crucial to achieving desirable outcomes such as higher material removal rate (MRR), improved surface roughness (Ra), better dimensional accuracy, and reduced tool wear. In recent years, numerous optimization techniques including Taguchi method, Grey Relational Analysis (GRA), Response Surface Methodology (RSM), Artificial Neural Networks (ANN), and Genetic Algorithms (GA) have been applied to enhance machining performance.

This review paper presents a comprehensive analysis of the WEDM process, focusing on process parameters, performance measures, and various optimization techniques reported in the literature. It also highlights key findings, compares different optimization approaches, and identifies existing research gaps. Furthermore, the study outlines future research directions such as the integration of artificial intelligence and hybrid optimization methods for improved machining efficiency and accuracy.

1. Introduction

In modern manufacturing industries, the demand for high precision, complex geometries, and machining of hard-to-cut materials has significantly increased (Alam et al., 2022; Rubi et al.,

2024). Conventional machining processes often face limitations when dealing with materials such as tool steels, superalloys, and composites due to high cutting forces, tool wear, and thermal damage (Nair *et al.*, 2022; Bhanage & Mulla, 2022). To overcome these challenges, non-traditional machining processes have gained considerable importance, among which Electrical Discharge Machining (EDM) is one of the most widely used techniques (Deshmukh *et al.*, 2019; Divya *et al.*, 2020).

Wire Electrical Discharge Machining (WEDM) is an advanced form of EDM that utilizes a continuously moving thin wire electrode to cut electrically conductive materials with high accuracy and excellent surface finish (Alam *et al.*, 2022; Juliyana *et al.*, 2024). The process is based on the principle of thermo-electric spark erosion, where material removal occurs due to a series of rapid electrical discharges between the wire electrode and the workpiece in the presence of a dielectric fluid (Kumar *et al.*, 2018; Mao *et al.*, 2025). Since there is no direct contact between the tool and the workpiece, WEDM eliminates mechanical stresses, making it highly suitable for machining delicate and complex components (Rubi *et al.*, 2024; Afridi *et al.*, 2025).

The performance of the WEDM process is strongly influenced by various machining parameters such as pulse-on time, pulse-off time, discharge current, wire feed rate, wire tension, and dielectric conditions (Naeim *et al.*, 2023; Natarajan *et al.*, 2022). These parameters directly affect important performance measures including material removal rate (MRR), surface roughness (Ra), kerf width, dimensional accuracy, and tool wear (Nair *et al.*, 2022; Radhakrishnan & Kesavalu, 2025). Improper selection of these parameters can lead to reduced machining efficiency, poor surface quality, and instability in the machining process (Kesavalu & Subramanian, 2024).

To achieve optimal machining performance, researchers have applied various optimization techniques such as the Taguchi method, Grey Relational Analysis (GRA), Response Surface Methodology (RSM), Artificial Neural Networks (ANN), and Genetic Algorithms (GA) (Datta & Mahapatra, 2010; Durairaj *et al.*, 2013; Kumar *et al.*, 2018; Chanie *et al.*, 2025). These methods help in identifying the most significant parameters and determining the optimal combination of machining conditions for improved productivity and quality (Pendokhare & Chakraborty, 2024). This review paper aims to provide a comprehensive overview of WEDM process parameters, performance characteristics, and optimization techniques. It critically analyzes previous research work, highlights key findings, and identifies research gaps for future investigation. The study also emphasizes the importance of advanced optimization and intelligent techniques in enhancing the

efficiency and reliability of the WEDM process in modern manufacturing applications (*Alam et al., 2022; Rubi et al., 2024*).

2. Literature Review

Wire Electrical Discharge Machining (WEDM) has been extensively studied by researchers to improve machining performance in terms of material removal rate (MRR), surface roughness (Ra), kerf width, and dimensional accuracy (*Alam et al., 2022; Rubi et al., 2024*). Various experimental, statistical, and intelligent optimization techniques have been applied to analyze and optimize the machining parameters (*Deshmukh et al., 2019; Divya et al., 2020*). The following section presents a comprehensive review of relevant literature.

Several researchers have focused on understanding the influence of process parameters on WEDM performance. Abdallah et al. (2021) investigated the effect of cut direction and machining parameters on carbon fibre-reinforced plastic composites and reported that fibre orientation significantly affects surface integrity and machining efficiency (*Abdallah et al., 2021*). Similarly, Naeim et al. (2023) examined WEDM of stainless steel 304 and found that pulse-on time and discharge current increase MRR but adversely affect surface finish due to higher thermal energy input (*Naeim et al., 2023*).

Experimental investigations on different materials have also been widely reported. Nair et al. (2022) studied WEDM of Inconel 718 and observed that higher discharge energy improves MRR but negatively impacts dimensional accuracy and kerf width (*Nair et al., 2022*). Bhanage and Mulla (2022) conducted experiments on AISI D3 tool steel and concluded that pulse-on time and current are the most influential parameters affecting both MRR and surface roughness (*Bhanage & Mulla, 2022*). Likewise, Radhakrishnan and Kesavalu (2025) analyzed duplex stainless steel and highlighted the trade-off between productivity and surface quality, emphasizing the need for parameter optimization (*Radhakrishnan & Kesavalu, 2025*).

A significant amount of research has been dedicated to optimization techniques. Datta and Mahapatra (2010) applied response surface methodology (RSM) combined with grey-Taguchi techniques for multi-objective optimization and demonstrated improved machining performance (*Datta & Mahapatra, 2010*). Kumar et al. (2018) used grey-based RSM to develop predictive models and optimize machining parameters effectively (*Kumar et al., 2018*). Similarly, Durairaj et al. (2013) integrated the Taguchi method with grey relational analysis (GRA) to achieve simultaneous optimization of multiple performance measures (*Durairaj et al., 2013*).

In recent years, advanced and hybrid optimization approaches have gained attention. Chanie et al. (2025) utilized artificial neural networks (ANN) and genetic algorithms (GA) for optimizing WEDM parameters and reported improved prediction accuracy and performance (*Chanie et al., 2025*). Afridi et al. (2025) also emphasized multi-objective optimization techniques for machining D2 steel, highlighting their effectiveness in balancing MRR and surface quality (*Afridi et al., 2025*).

Review studies have provided broader insights into WEDM research trends. Alam et al. (2022) and Rubi et al. (2024) presented comprehensive reviews on WEDM performance evaluation and optimization techniques, identifying key parameters and challenges such as recast layer formation, thermal damage, and wire breakage (*Alam et al., 2022; Rubi et al., 2024*). Pendokhare and Chakraborty (2024) reviewed multi-objective optimization methods and concluded that hybrid and AI-based techniques offer superior performance for complex machining problems (*Pendokhare & Chakraborty, 2024*). Additionally, Divya et al. (2020) and Deshmukh et al. (2019) highlighted the effectiveness of grey relational analysis in handling multi-response optimization problems (*Divya et al., 2020; Deshmukh et al., 2019*).

Furthermore, recent studies such as Mao et al. (2025) demonstrated the application of grey relational analysis for optimizing machining of advanced materials like 9Cr18Mov steel (*Mao et al., 2025*), while Juliyana et al. (2024) validated the effectiveness of the Taguchi method in improving machining performance for aluminium alloys (*Juliyana et al., 2024*). Kesavalu and Subramanian (2024) also emphasized the importance of surface integrity aspects, including micro-cracks and recast layer thickness, in addition to conventional performance measures (*Kesavalu & Subramanian, 2024*).

From the literature, it is evident that pulse-on time, pulse-off time, and discharge current are the most significant parameters influencing WEDM performance (*Natarajan et al., 2022; Naeim et al., 2023*). While traditional optimization techniques such as Taguchi and GRA are widely used due to their simplicity, modern approaches involving artificial intelligence and hybrid models are increasingly being adopted for more accurate and efficient optimization (*Pendokhare & Chakraborty, 2024; Chanie et al., 2025*).

Table 1 Materials and key parameters

Sr. No.	Author & Year	Material Used	Method/Technique	Input Parameters	Output Responses	Key Findings
1	Abdallah et al. (2021)	CFRP Composite	Experimental	Pulse time, current, speed, wire	MRR, Surface Integrity	Cut direction significantly affects machining performance
2	Alam et al. (2022)	Various	Review	Multiple parameters	MRR, Ra, Kerf width	Highlighted importance of optimization techniques
3	Bhanage & Mulla (2022)	AISI D3 Steel	Experimental + Optimization	Ton, Toff, Current	MRR, Ra	Pulse-on time & current most influential
4	Chanie et al. (2025)	AISI 1020 Steel	ANN + GA	Ton, Toff, Current	MRR, Ra	Hybrid AI approach improves optimization accuracy
5	Datta & Mahapatra (2010)	Steel	RSM + Grey Taguchi	Ton, Toff, Current	MRR, Ra	Effective multi-objective optimization
6	Deshmukh et al. (2019)	Various	Review (Taguchi + GRA)	Multiple	MRR, Ra	GRA useful for multi-response optimization
7	Divya et al. (2020)	Various	Review	Pulse, current, tension	MRR, Ra	GRA widely used technique
8	Durairaj et al. (2013)	Stainless Steel	Taguchi + GRA	Ton, Toff, Current	MRR, Ra, Kerf	Combined approach improves performance
9	Juliyana et al. (2024)	LM5 Aluminium	Taguchi	Ton, Toff, Current	MRR, Ra	Taguchi effective for parameter optimization
10	Kesavalu & Subramanian (2024)	Duplex SS	Experimental	Ton, Toff, Current	MRR, Surface Integrity	Surface defects depend on parameter selection
11	Kumar et al. (2018)	Steel	Grey-based RSM	Ton, Toff, Current	MRR, Ra	Hybrid statistical model improves prediction
12	Mao et al. (2025)	9Cr18Mov Steel	GRA	Ton, Toff, Current	MRR, Ra	GRA effective for multi-objective optimization
13	Naeim et al. (2023)	SS304	Experimental	Ton, Toff, Current	MRR, Ra	Higher energy increases MRR but reduces surface quality
14	Nair et al. (2022)	Inconel 718	Experimental	Ton, Toff, Current	MRR, Kerf, Accuracy	Trade-off between MRR and accuracy
15	Natarajan et	Stainless Steel	Optimization	Ton, Toff,	MRR, Ra	Pulse-on time

	al. (2022)			Current		dominant factor
16	Pendokhare & Chakraborty (2024)	Various	Review	Multiple	MRR, Ra	AI and hybrid methods are future trend
17	Radhakrishnan & Kesavalu (2025)	Duplex SS	Experimental	Ton, Toff, Current	MRR, Ra, Tool Wear	Higher current increases tool wear
18	Rubi et al. (2024)	Various	Review	Multiple	MRR, Ra	Identified challenges like recast layer

3. Process Parameters in WEDM

The performance of the Wire Electrical Discharge Machining (WEDM) process is significantly influenced by a number of process parameters that control spark generation, energy distribution, and material removal mechanisms. These parameters must be carefully selected and optimized to achieve desirable machining characteristics such as higher material removal rate (MRR), improved surface roughness (Ra), and better dimensional accuracy. Broadly, WEDM parameters can be classified into electrical and non-electrical categories.

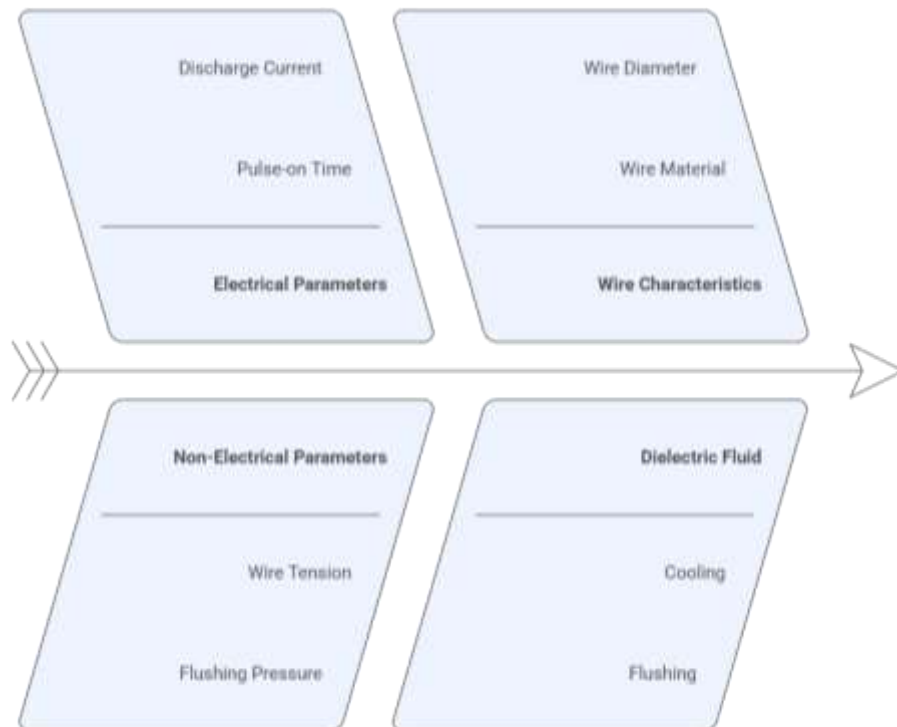


Figure 1: Optimizing WEDM Performance

3.1 Electrical Parameters

Electrical parameters directly control the discharge energy and spark behavior during the

machining process. Among these, pulse-on time (T_{on}) is one of the most critical factors, as it determines the duration of spark generation. An increase in pulse-on time leads to higher discharge energy, which enhances the material removal rate. However, excessive pulse-on time may result in poor surface finish due to the formation of larger craters and increased thermal damage.

Pulse-off time (T_{off}) represents the interval between successive sparks and plays an important role in stabilizing the machining process. It allows sufficient time for debris removal and dielectric recovery. A higher pulse-off time improves surface quality but reduces the frequency of spark generation, thereby lowering the material removal rate.

Discharge current is another significant parameter that governs the intensity of the spark. Higher current values increase the amount of heat generated, which improves material removal but simultaneously deteriorates surface quality and increases the likelihood of wire wear or breakage. Similarly, the applied voltage affects the spark gap and machining stability. Higher voltage generally results in a larger spark gap and stable machining conditions, but it may also contribute to rougher surfaces.

The duty cycle, defined as the ratio of pulse-on time to the total cycle time, reflects the proportion of active machining time. A higher duty cycle increases energy input and material removal rate, whereas a lower duty cycle improves surface finish.

3.2 Non-Electrical Parameters

Non-electrical parameters influence the mechanical stability and flushing efficiency of the WEDM process. The wire feed rate determines the speed at which fresh wire is supplied to the machining zone. An adequate wire feed rate helps reduce wire wear and maintain consistent machining performance.

Wire tension is another important factor that ensures the straightness and stability of the wire electrode. Proper tension improves dimensional accuracy and prevents wire vibration, although excessive tension may lead to wire breakage.

The dielectric fluid, typically deionized water, plays a crucial role in cooling the machining zone, removing debris, and maintaining electrical insulation until breakdown occurs. Effective flushing of debris is essential for stable machining, and this is controlled by flushing pressure. Inadequate flushing can lead to accumulation of debris, resulting in unstable sparking and poor surface finish. Additionally, the material and diameter of the wire electrode influence machining performance. Thinner wires provide better precision and finer cuts, while thicker wires allow higher material

removal rates but may compromise accuracy.

3.3 Influence of Parameters on Machining Performance

The combined effect of electrical and non-electrical parameters determines the overall efficiency and quality of the WEDM process. Higher discharge energy, achieved through increased pulse-on time and current, generally improves material removal rate but adversely affects surface finish and tool life. On the other hand, appropriate selection of pulse-off time, wire tension, and dielectric conditions enhances surface quality and machining stability.

Overall, it can be concluded that pulse-on time, pulse-off time, and discharge current are the most influential parameters in WEDM. Their proper optimization is essential to achieve a balance between productivity and surface integrity, making parameter selection a critical aspect of WEDM process control.

4. PERFORMANCE MEASURES

The performance of the Wire Electrical Discharge Machining (WEDM) process is evaluated based on several output characteristics that reflect machining efficiency, quality, and accuracy. These performance measures are strongly influenced by process parameters and are essential for assessing and optimizing the machining process. Among these, material removal rate (MRR), surface roughness (Ra), kerf width, tool wear rate, and dimensional accuracy are the most commonly studied responses.

4.1 Material Removal Rate (MRR)

Material Removal Rate (MRR) is one of the most important indicators of machining productivity. It represents the volume of material removed from the workpiece per unit time. Higher MRR is generally desirable as it reduces machining time and improves efficiency. In WEDM, MRR is primarily influenced by discharge energy, which depends on parameters such as pulse-on time and discharge current. An increase in these parameters leads to higher thermal energy generation, resulting in faster material removal. However, excessive energy input may compromise surface quality and lead to instability in the machining process.

4.2 Surface Roughness (Ra)

Surface roughness is a critical measure of the quality of the machined surface. It indicates the degree of irregularities and texture formed on the workpiece due to spark erosion. In WEDM, surface roughness is largely affected by discharge energy and flushing conditions. Higher pulse-on time and current tend to produce deeper and larger craters, resulting in a rougher surface.

Conversely, appropriate control of pulse-off time and dielectric conditions helps in achieving smoother surfaces. Surface roughness is particularly important in applications requiring high precision and superior surface finish.

4.3 Kerf Width

Kerf width refers to the width of the cut produced during the machining process. It is an important parameter in determining dimensional accuracy and material loss. Kerf width depends on factors such as wire diameter, spark gap, and discharge energy. Higher discharge energy increases the erosion zone, leading to wider kerf. Accurate control of kerf width is essential for maintaining tight tolerances in precision components.

4.4 Tool Wear Rate (TWR)

Although WEDM uses a continuously moving wire electrode, tool wear still occurs in the form of wire degradation and breakage. Tool wear rate affects machining stability, accuracy, and cost. High discharge energy and improper flushing conditions can accelerate wire wear. Therefore, maintaining optimal machining conditions is necessary to minimize tool wear and ensure consistent performance.

4.5-Dimensional Accuracy

Dimensional accuracy is a measure of how closely the machined component matches the desired dimensions. It is influenced by several factors including wire tension, spark gap, thermal effects, and machine stability. In WEDM, maintaining high dimensional accuracy is essential for producing precision components used in aerospace, automotive, and tooling industries. Improper parameter settings can lead to overcut, tapering, and geometric inaccuracies.

4.6 Summary

In WEDM, performance measures such as MRR, surface roughness, kerf width, tool wear rate, and dimensional accuracy are interrelated and often involve trade-offs. For instance, increasing MRR may result in poorer surface finish and reduced accuracy. Therefore, it is necessary to optimize process parameters in such a way that a balance is achieved between productivity and quality.

5. OPTIMIZATION TECHNIQUES

Optimization of process parameters in Wire Electrical Discharge Machining (WEDM) is essential to achieve the desired balance between productivity and quality. Since multiple input parameters simultaneously influence several performance measures such as material removal rate (MRR),

surface roughness (Ra), and dimensional accuracy, the selection of optimal machining conditions becomes a complex task. To address this challenge, various statistical, mathematical, and intelligent optimization techniques have been developed and widely applied in WEDM research.

5.1 Taguchi Method

The Taguchi method is one of the most commonly used statistical techniques for process optimization in WEDM due to its simplicity and effectiveness. It is based on the design of experiments (DOE) using orthogonal arrays, which significantly reduces the number of experimental runs while still providing reliable results. The method uses signal-to-noise (S/N) ratios to evaluate the performance characteristics and identify the optimal levels of process parameters. Depending on the objective, different S/N criteria such as “larger is better” for MRR and “smaller is better” for surface roughness are applied. The Taguchi method also incorporates analysis of variance (ANOVA) to determine the significance of each parameter. Although it is highly efficient for single-response optimization, its capability is limited when dealing with multiple performance characteristics.

5.2 Grey Relational Analysis (GRA)

Grey Relational Analysis is widely used for multi-objective optimization in WEDM processes. It is particularly effective when multiple performance measures need to be optimized simultaneously. In this method, experimental results are first normalized and then converted into a grey relational coefficient, which represents the relationship between the ideal and actual performance. These coefficients are then combined to obtain a grey relational grade, which serves as a single performance index. By analyzing this grade, optimal parameter settings can be identified. GRA is often combined with the Taguchi method to enhance its multi-response optimization capability.

5.3 Response Surface Methodology (RSM)

Response Surface Methodology is a statistical and mathematical technique used to develop empirical models and analyze the relationship between input parameters and output responses. In WEDM, RSM helps in constructing regression models that predict machining performance under different conditions. It also enables the generation of contour plots and response surfaces, which provide a visual understanding of parameter interactions. RSM is particularly useful for identifying optimal regions and understanding the combined effects of multiple variables. However, it requires more experimental data compared to the Taguchi method.

5.4 Artificial Neural Network (ANN)

Artificial Neural Networks are intelligent modeling tools inspired by the human brain, capable of learning complex nonlinear relationships between input and output variables. In WEDM, ANN is used to predict machining performance based on training data obtained from experiments. Once trained, the model can accurately estimate responses such as MRR and surface roughness for different parameter combinations. ANN is highly effective for complex systems, but its performance depends on the quality and quantity of training data.

5.5 Genetic Algorithm (GA)

Genetic Algorithm is an evolutionary optimization technique based on the principles of natural selection and genetics. It is used to find optimal solutions by iteratively improving a population of candidate solutions. In WEDM, GA is often combined with predictive models such as ANN or RSM to determine the best combination of process parameters. It is particularly useful for solving complex multi-objective optimization problems where traditional methods may not be effective.

5.6 Hybrid Optimization Techniques

In recent years, hybrid optimization approaches that combine multiple techniques have gained significant attention. Methods such as Taguchi-GRA, ANN-GA, and RSM-GRA integrate the advantages of different approaches to achieve more accurate and reliable optimization results. These hybrid techniques are especially effective in handling multi-objective problems and complex machining environments, making them highly suitable for advanced WEDM applications.

Overall, optimization techniques play a crucial role in enhancing WEDM performance. While traditional methods such as Taguchi and GRA are widely used due to their simplicity, advanced techniques like ANN and GA offer higher accuracy and better handling of complex relationships. The selection of an appropriate optimization method depends on the nature of the problem, the number of responses involved, and the desired level of accuracy.

6. Conclusion

In conclusion, Wire Electrical Discharge Machining (WEDM) has emerged as a highly effective and versatile non-traditional machining process for manufacturing complex and high-precision components from electrically conductive materials. Its ability to machine hard-to-cut materials with excellent accuracy, minimal mechanical stress, and superior surface finish makes it indispensable in modern industries such as aerospace, automotive, and tooling.

The review highlights that the performance of WEDM is significantly influenced by key process parameters, particularly pulse-on time, pulse-off time, and discharge current. These parameters directly affect critical performance measures such as material removal rate (MRR), surface roughness (Ra), kerf width, and dimensional accuracy. A major observation across the literature is the inherent trade-off between productivity and surface quality, where higher discharge energy improves MRR but may deteriorate surface integrity and accuracy.

Various optimization techniques have been extensively applied to address this challenge. Traditional methods like the Taguchi method and Grey Relational Analysis (GRA) are widely used due to their simplicity and efficiency, while advanced approaches such as Response Surface Methodology (RSM), Artificial Neural Networks (ANN), and Genetic Algorithms (GA) provide more accurate and reliable optimization for complex, multi-objective problems. Hybrid optimization techniques combining these methods have shown superior performance in achieving balanced machining outcomes.

Furthermore, the review identifies several research gaps, including the need for deeper investigation into surface integrity issues such as recast layer formation, micro-cracks, and thermal damage, as well as improved understanding of wire breakage mechanisms. Future research should focus on the integration of artificial intelligence, machine learning, and hybrid optimization strategies to enhance prediction accuracy and process control.

Overall, effective optimization of WEDM parameters is essential to achieve a balance between machining efficiency and product quality. The continuous advancement of intelligent and hybrid optimization techniques is expected to further improve the performance, reliability, and industrial applicability of the WEDM process.

7. Conflict of interest

None

8. References

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